Lab 1: Requirements Engineering & UML Use-Case Modelling

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Problem statement 11: Telemedicine Slot Booking & Prescription Portal

The system enables patients to find doctors, book available video-consultation slots, receive a secure consultation-room link and access digitally signed prescriptions. This submission focuses on reliable booking, secure consultation access and verifiable prescription records.

1. Functional requirements

ID	Type	Description	Priority	Acceptance criteria	Rationale
FR-001	Functional	The system shall allow patients to view active doctors and their specialties before selecting a video consultation.	High	Pass: Active doctor profiles display specialty information before booking. Fail: an inactive or unavailable doctor can be selected for a new booking.	Enables informed doctor selection and prevents booking against unavailable physician profiles.
FR-002	Functional	The system shall allow a patient to select and book an available video-consultation date and time slot for a selected doctor.	High	Pass: Only currently available slots can be selected and a successful booking returns a confirmation with doctor, date and time. Fail: a slot already booked by another patient is confirmed.	Provides the core scheduling function and prevents double booking.
FR-003	Functional	The system shall generate a secure consultation-room link only after a booking is confirmed and make the link available to the patient for the scheduled consultation.	High	Pass: A confirmed booking receives a room link and the link is visible through the authorized patient portal. Fail: an unconfirmed booking exposes a consultation link.	Connects a valid booking to a controlled remote-consultation entry point.

Requirements Table (continued)

ID	Type	Description	Priority	Acceptance criteria	Rationale
FR-004	Functional	The system shall allow an attending physician to author a structured prescription and digitally sign it after successful physician-license validation.	High	Pass: License validation succeeds before signing and the generated prescription PDF contains a valid digital signature. Fail: a prescription is signed without successful license validation.	Makes the prescription attributable and verifiable.
FR-005	Functional	The system shall allow an authorized patient to download the digitally signed prescription as a PDF.	Medium	Pass: The patient can download the prescription as a PDF and the downloaded record retains the physician signature. Fail: an unsigned record is presented as a signed prescription.	Provides a portable medical record while preserving authenticity.
NFR-001	Performance & availability	The consultation-booking service shall respond to a booking request within 2 seconds under the agreed normal load and maintain at least 99.9% monthly availability.	High	Pass: Load testing meets the 2-second booking-response target and measured monthly availability is at least 99.9%. Fail: either target is missed.	Keeps appointment booking responsive and dependable.
NFR-002	Security & auditability	The system shall encrypt patient and prescription data in transit and at rest, restrict access by authenticated identity and record security-relevant booking and prescription events in an audit trail.	High	Pass: Test connections use TLS 1.2 or later, stored sensitive data uses AES-256, unauthorized access is denied, and tested security events record time, actor and relevant record identifiers. Fail: encryption, access control or audit logging is missing.	Protects sensitive healthcare information and supports investigation of security events.

Test settings: The response-time and availability values above are proposed verification targets for this lab submission, not clinical recommendations. Operational limits and access policies are configurable.

Requirement IDs refer to this individual submission.

2. UML Use-Case Diagram

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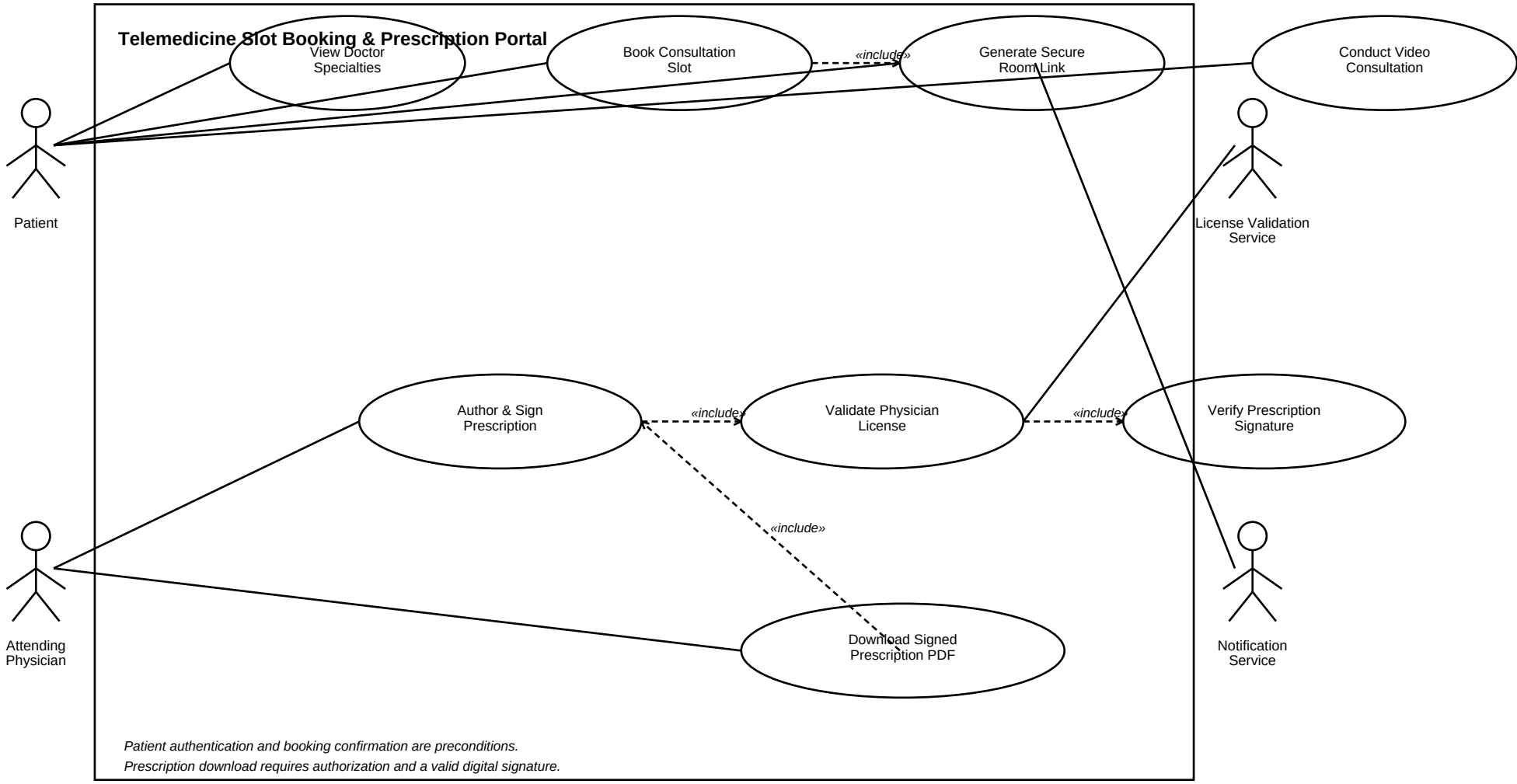

3. Use-Case Flow Specification

UC-02: Book Video Consultation Slot

Primary actor: Patient

Other participants: Attending Physician / Scheduling Service; Notification Service

Trigger: A patient chooses a doctor and requests an available video-consultation slot.

Related requirements: FR-001 to FR-005; NFR-001 and NFR-002.

Preconditions

1. The patient is registered and authenticated.
2. At least one active physician and one available consultation slot exist.
3. The scheduling and notification services are available.
4. The patient has permission to create a consultation booking.

Main success scenario

1. The patient opens the telemedicine portal and views available doctors and specialties.
2. The patient selects an active physician and chooses Book Video Consultation Slot.
3. The system displays currently available dates and times for the selected physician.
4. The patient selects an available slot and submits the booking request.
5. The system validates the patient identity, physician availability and selected slot.
6. The system reserves the slot and creates a confirmed booking.
7. The system generates a secure consultation-room link for the confirmed booking.
8. The system sends the booking confirmation and room link through the configured notification channel.
9. The patient can use the secure link at the scheduled time to conduct the video consultation.
10. The system records the booking and relevant access events in the audit trail.

Alternate flow A1: Selected slot becomes unavailable

A1.1 The selected slot is taken by another patient before confirmation.

A1.2 The system rejects the booking and informs the patient that the slot is no longer available.

A1.3 The system displays other currently available slots.

A1.4 The patient selects another slot and resumes at step 4.

A1.5 If no slots remain, the use case ends without a booking.

Postconditions

Success: A valid consultation booking is stored, the selected slot is reserved, and a secure room link is generated and delivered.

Alternate outcome: No booking is created for an unavailable slot; the patient is offered other available slots.